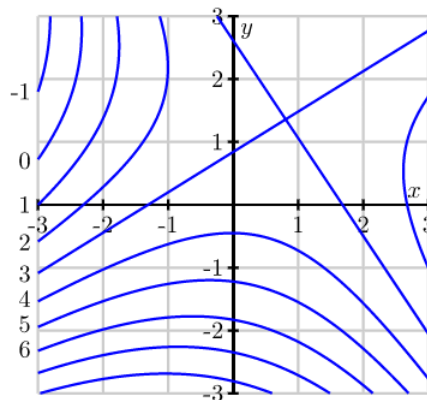


Math 2551 Worksheet: Partial Derivatives

1. Below is a contour plot for a function $f(x, y)$, with values for some of the contours indicated on the left of the figure.

- (a) Estimate the partial derivative $f_x(-2, -1)$.
- (b) Estimate the partial derivative $f_y(-2, -1)$.
- (c) Locate, if possible, one point (x, y) where $f_x(x, y) = 0$.
- (d) Locate, if possible, one point (x, y) where $f_x(x, y) < 0$.



2. The speed of sound C traveling through ocean water is a function of temperature, salinity, and depth. It may be modeled by the function

$$C(T, S, D) = 1450 + 4.5T - 0.05T^2 + 0.0003T^3 + (1.5 - 0.01T)(S - 35) + 0.015D,$$

where C is the speed of sound in meters/second, T is the temperature in degrees Celsius, S is the salinity in grams/liter of water, and D is the depth below the ocean surface in meters.

- (a) State the units in which each of the partial derivatives C_T , C_S , and C_D are expressed and explain the physical meaning of each.
 - (b) Find the partial derivatives C_T , C_S , and C_D .
 - (c) Evaluate each of the three partial derivatives at the point where $T = 10$, $S = 35$, and $D = 100$. What does the sign of each partial derivative tell us about the behavior of the function C at the point $(10, 35, 100)$?
3. Find all the second partial derivatives for $f(x, y) = e^x + x \ln(y)$.
4. Find f_x , f_y , f_z , and f_{xzz} for the function $f(x, y, z) = x \sin(yz)$.

5. Find the total derivative Df at the given point for each function below. Remember that Df is the matrix of (partial) derivatives of the function and if f is a function from $\mathbb{R}^n$ to $\mathbb{R}^m$ then Df is a $m \times n$ matrix.

(a) $f(x) = 2x^3 + 7$ at $x = 2$.

(b) $\mathbf{f}(t) = \langle 2\cos(t), 2\sin(t), t \rangle$ at $t = \pi/2$.

(c) $f(x, y) = \sqrt{y - x}$ at $(x, y) = (1, 2)$.

(d) $f(x, y, z) = e^{2y-x} + z^2 + 4$ at $(x, y, z) = (1, 2, 3)$.

(e) $\mathbf{f}(s, t) = \langle 2s + 3t, t - s \rangle$ at $(s, t) = (1, 1)$.

Note: The graph of this function is a surface (in this case all of $\mathbb{R}^2$) parameterized by two variables just like the graph of the function in (b) is a curve parameterized by one variable - we'll see these more later! Another way of thinking about this is that this is a *change of variables* for $\mathbb{R}^2$ between the system of coordinates (s, t) and (x, y) .

6. The fifth-order partial derivative $\partial^5 f / \partial x^2 \partial y^3$ is zero for each of the following functions. To show this as quickly as possible, which variable would you differentiate with respect to first: x or y ?

Try to answer without writing anything down. Why did you make the choice you did?

(a) $f(x, y) = y^2 x^4 e^x + 2$

(b) $f(x, y) = y^2 + y(\sin(x) - x^4)$

(c) $f(x, y) = x^2 + 5xy + \sin(x) + 7e^x$

(d) $f(x, y) = x e^{y/2}$